Akoizo Original Practice - 2027 Rotation Form 2 - Answer key

Division C | Anatomy And Physiology | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Type II pneumocytes produce:

Unit: Lungs, Alveoli, Pleura, and Breathing Muscles.
Worked-case data: At one instant, alveolar pressure is 0 cm H_2O relative to atmosphere and pleural pressure is -7 cm H_2O. At another, alveolar pressure is -1 and pleural pressure is -6.

A. HCl

B. Antibodies

C. Mucus

D. Surfactant

Correct: D

2. (2 points) Exercising muscle unloads more O2 because of:

Unit: Gas Exchange, Hemoglobin, and pH Control.
Worked-case data: Compare blood with hemoglobin 15 g/dL versus 8 g/dL. Both samples have saturation 0.98 and PO_2 100 mmHg.

A. High pH

B. Left shift from alkalosis

C. Right shift from acid, CO2, heat

D. Low 2,3-BPG

Correct: C

3. (2 points) Apical cavitary disease with caseating granulomas suggests:

Unit: Infection and Restriction: Pneumonia, TB, Fibrosis.
Worked-case data: Case A has a focal air-space opacity and ongoing blood flow through poorly ventilated tissue. Case B has diffuse interstitial thickening and reduced total lung capacity.

A. Asthma

B. TB

C. Emphysema

D. Pneumothorax

Correct: B

4. (2 points) B12 is absorbed in the:

Unit: Small Intestine, Liver, Pancreas, and Absorption.
Worked-case data: A case supplies normal pancreatic enzyme production but greatly reduced bile delivery to the small intestine. Predict the most directly affected step.

A. Jejunum

B. Terminal ileum

C. Colon

D. Duodenum

Correct: B

5. (2 points) Lactose intolerance is caused by deficiency of:

Unit: Lactose Intolerance, Obesity, and Exercise Metabolism.
Worked-case data: A teaching model supplies 24 g lactose during transit. In one setting, 18 g can be hydrolyzed; in another, only 6 g can be hydrolyzed.

A. Maltase

B. Pepsin

C. Sucrase

D. Lactase

Correct: D

6. (2 points) Hard fixed nontender supraclavicular node most suggests:

Unit: Spleen, Nodes, and Barriers in Defense.
Worked-case data: One case involves particles arriving from a skin wound in lymph. Another involves microbes circulating in blood. A third involves intestinal luminal exposure.
Separate investigation observations:
Input: The organ receives and filters blood.
Architecture: White pulp supports immune responses; red pulp processes erythrocytes.
Function: The case emphasizes defense against blood-borne material.

A. Vaccine reaction

B. Viral URI

C. Malignancy

D. Allergy

Correct: C

7. (2 points) CD8 T cells recognize antigen on:

Unit: Adaptive Immunity, Antibodies, and Allergy.
Worked-case data: An experiment shows a delayed small antibody rise after first exposure and an earlier larger rise after a later exposure to the same antigen.

A. MHC II only

B. MHC I

C. Free antibody

D. Complement

Correct: B

8. (2 points) Heat-worsened blurred vision with INO suggests:

Unit: Autoimmunity, MS, RA, and Systemic Allergy.
Worked-case data: Case A describes central myelin injury. Case B describes destructive synovial inflammation. Case C describes rapid systemic effects after re-exposure to an external antigen.
Separate investigation observations:
Timing: Effects develop rapidly after re-exposure.
Trigger: The trigger is an external antigen encountered previously.
Mechanism: Mast-cell mediators alter vascular permeability and airway smooth muscle.

A. MS

B. Gout

C. Celiac

D. RA

Correct: A

9. (2 points) Normal pleural pressure during quiet breathing is:

Unit: Lungs, Alveoli, Pleura, and Breathing Muscles.
Worked-case data: At one instant, alveolar pressure is 0 cm H_2O relative to atmosphere and pleural pressure is -7 cm H_2O. At another, alveolar pressure is -1 and pleural pressure is -6.

A. Negative around -4 mmHg

B. Equal to alveolar

C. Positive 10 mmHg

D. Atmospheric

Correct: A

10. (2 points) Most CO2 is carried as:

Unit: Gas Exchange, Hemoglobin, and pH Control.
Worked-case data: Compare blood with hemoglobin 15 g/dL versus 8 g/dL. Both samples have saturation 0.98 and PO_2 100 mmHg.

A. Myoglobin

B. Dissolved gas

C. Carbamino only

D. Bicarbonate

Correct: D

11. (2 points) A teaching spirometry case gives tidal volume 540 mL, dead space 150 mL and respiratory rate 12/min. Find alveolar ventilation.

answer: 4680 mL/min (2 points).

Numerical target 4680; absolute tolerance 9.36. Accepted units: , mL/min.

12. (2 points) Use the same case to calculate minute ventilation: tidal volume 540 mL and rate 12/min.

answer: 6480 mL/min (2 points).

Numerical target 6480; absolute tolerance 12.96. Accepted units: , mL/min.

13. (2 points) A model narrows an airway while pressure difference and other conditions are fixed. Explain the likely airflow consequence.

answer: Narrowing increases resistance and generally reduces flow for a fixed driving pressure; real airways also depend on compliance and flow regime. (2 points).

14. (2 points) Describe how pleural fluid assists ordinary breathing.

answer: It lubricates pleural surfaces so they can slide with low friction while the pleural coupling helps chest and lung motion interact. (2 points).

15. (2 points) Why does the small intestine combine villi, microvilli and a thin absorptive epithelium?

answer: They increase effective exchange area and support absorption while limiting transport distance, with blood/lymph pathways carrying absorbed products. (2 points).

16. (2 points) Distinguish the digestive roles of bile and pancreatic lipase.

answer: Bile salts aid emulsification and micellar handling of lipids; lipase enzymatically hydrolyzes triglycerides. Bile is not itself a lipase. (2 points).

17. (2 points) Explain why a pancreatic bicarbonate secretion can assist digestion of acidic stomach contents.

answer: It helps neutralize acid in the small intestine and supports an appropriate environment for intestinal and pancreatic enzyme activity. (2 points).

18. (2 points) Compare innate and adaptive immune responses without calling either one universally faster or stronger in every situation.

answer: Innate mechanisms provide broad early recognition and defense; adaptive lymphocyte responses are antigen-specific and can generate memory, with timing affected by prior exposure. (2 points).

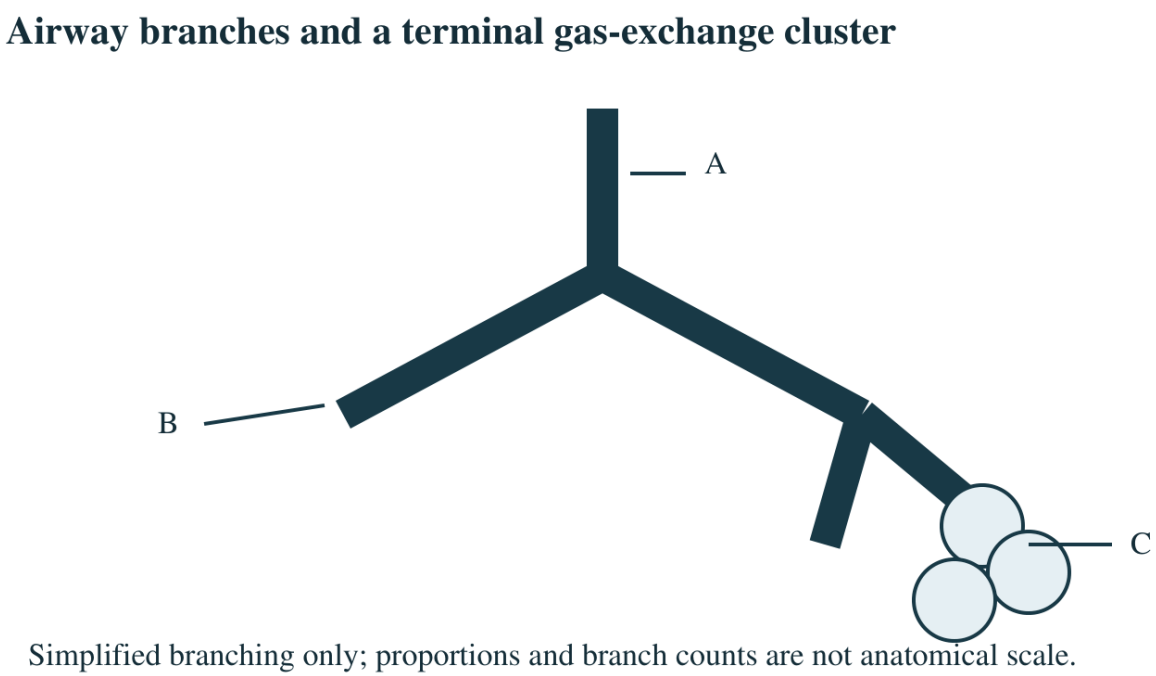
19. (2 points) Explain why a secondary antibody response can differ from a primary response.

answer: Memory B cells and other adaptive changes can produce a faster, larger and higher-affinity response after re-exposure to the same antigen. (2 points).

20. (2 points) In a teaching case, helper T-cell activation is impaired. Give one potential consequence for a coordinated adaptive response.

answer: Impaired helper signaling can reduce effective B-cell activation/class switching or other coordinated immune responses; the consequence depends on the pathway and context. (2 points).

21. (2 points) Use Figure F1. Name the structure labeled A.



F1. Respiratory branching schematic (original synthetic teaching figure)

answer: Trachea. (2 points).

22. (2 points) Use Figure F1. Which labeled region most directly provides the thin gas-exchange interface?

Refer to F1. Respiratory branching schematic (original synthetic teaching figure) shown with the first question in this figure group.

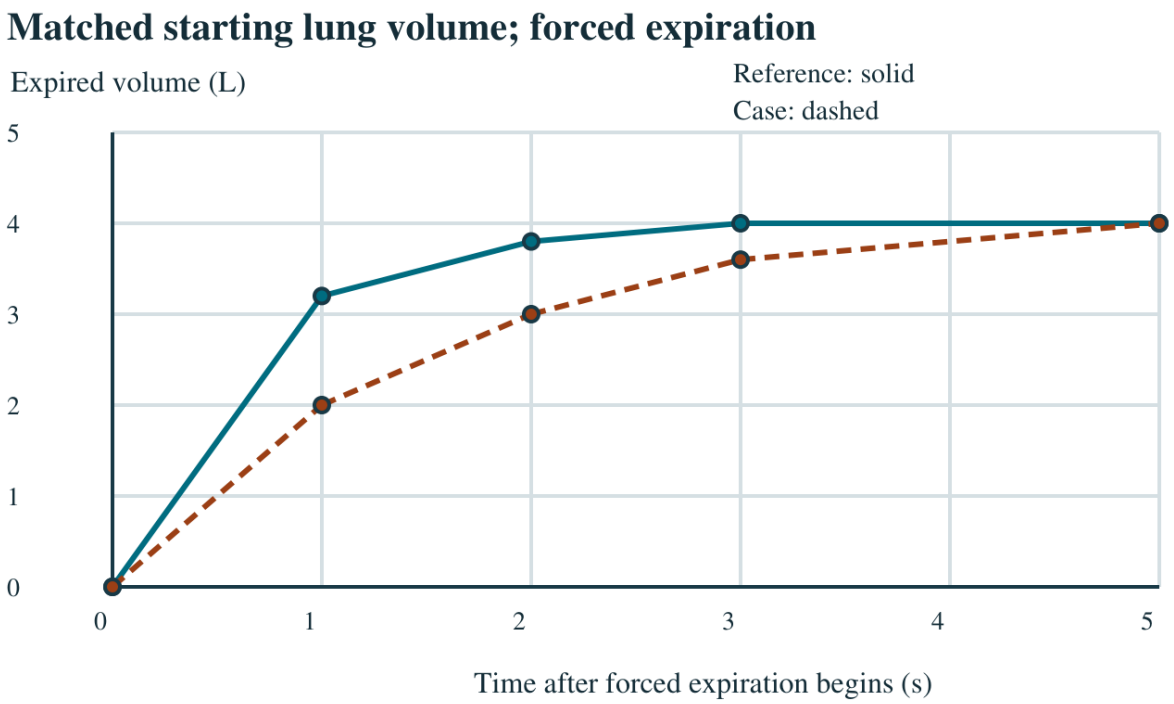
answer: C, the alveolar cluster. (2 points).

23. (2 points) Use Figure F1. Trace inspired air from A toward C using the branching arrangement.

Refer to F1. Respiratory branching schematic (original synthetic teaching figure) shown with the first question in this figure group.

answer: Air passes from trachea through a main bronchus and progressively smaller bronchi/bronchioles to alveoli; the drawing omits many intermediate branches. (2 points).

24. (2 points) Use Figure F2. Calculate case FEV1/FVC as a percentage.



F2. Forced expiratory volume curves (original synthetic teaching figure)

answer: 50 % (2 points).

Numerical target 50; absolute tolerance 0.15. Accepted units: , %.

25. (2 points) Use Figure F2. Which curve shows slower early emptying, and what pattern does that support studying?

Refer to F2. Forced expiratory volume curves (original synthetic teaching figure) shown with the first question in this figure group.

answer: The case/dashed curve; reduced early expired volume relative to final capacity supports studying an obstructive pattern, without establishing a specific disease by itself. (2 points).

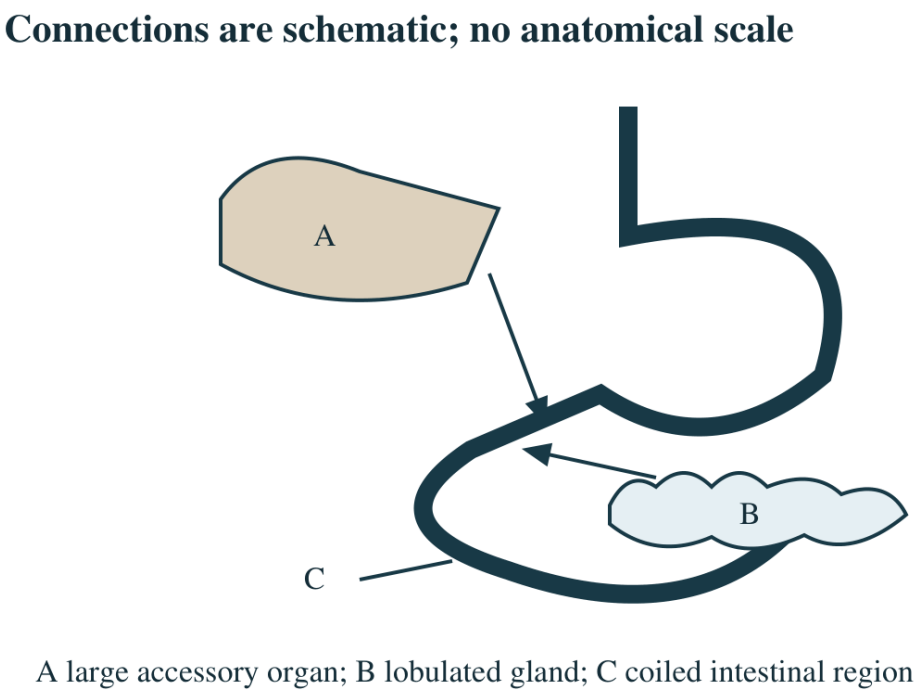
26. (2 points) Use Figure F2. Find reference-minus-case expired volume at1 s.

Refer to F2. Forced expiratory volume curves (original synthetic teaching figure) shown with the first question in this figure group.

answer: 1.2 L (2 points).

Numerical target 1.2; absolute tolerance 0.0036. Accepted units: , L.

27. (2 points) Use Figure F3. Which labeled organ produces bile? Give its name.



F3. Digestive organ relationships (original synthetic teaching figure)

answer: A, the liver. (2 points).

28. (2 points) Use Figure F3. Name B and explain one digestive secretion it delivers toward the shown intestinal connection.

Refer to F3. Digestive organ relationships (original synthetic teaching figure) shown with the first question in this figure group.

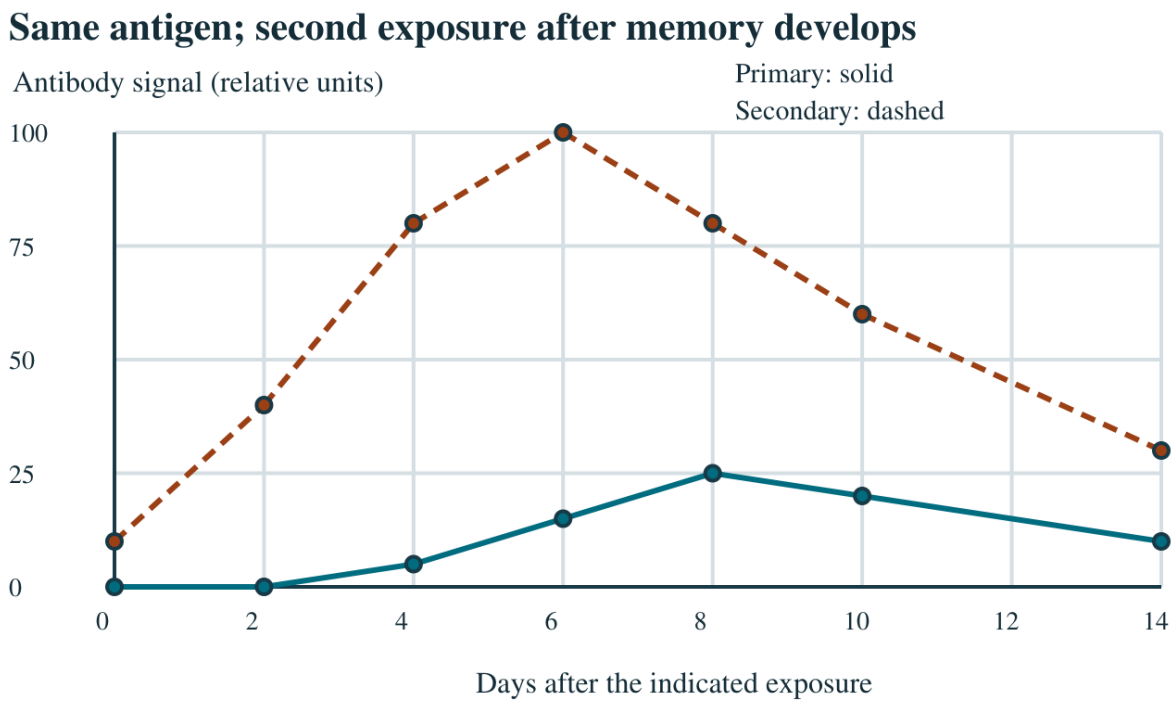
answer: Pancreas; digestive enzymes and bicarbonate enter the duodenum through pancreatic ducts. (2 points).

29. (2 points) Use Figure F3. Which labeled region is most directly associated with most nutrient absorption?

Refer to F3. Digestive organ relationships (original synthetic teaching figure) shown with the first question in this figure group.

answer: C, the small-intestinal region. (2 points).

30. (2 points) Use Figure F4. Read the ratio secondary-peak / primary-peak signal.



F4. Primary and secondary antibody responses (original synthetic teaching figure)

answer: 4 (2 points).

Numerical target 4; absolute tolerance 0.012. Accepted units: .

31. (2 points) Use Figure F4. At day4, calculate secondary-minus-primary signal.

Refer to F4. Primary and secondary antibody responses (original synthetic teaching figure) shown with the first question in this figure group.

answer: 75 relative units (2 points).

Numerical target 75; absolute tolerance 0.225. Accepted units: , relative units.

32. (2 points) Use Figure F4. Explain how the curve comparison is consistent with adaptive immune memory.

Refer to F4. Primary and secondary antibody responses (original synthetic teaching figure) shown with the first question in this figure group.

answer: The secondary response is faster and larger, consistent with memory lymphocyte activation; the schematic alone does not identify every cellular mechanism. (2 points).